Java Fundamentals - Exercises

Java Introduction & Features

- - What is Java and who developed it?
- - List any 5 features of Java.
- - Why is Java called platform-independent?

JVM, JRE, JDK - Architecture

- - Differentiate between JVM, JRE, and JDK.
- - Draw and explain the architecture of JVM.
- - What role does the JIT compiler play?

Java Program Structure

- - Write a basic Java program to print 'Hello, Java!'.
- - Explain each part of a Java program.
- - What is the role of the 'main' method?

Keywords and Identifiers

- - Define keywords with examples.
- - List 10 Java keywords.
- - What are identifiers and what rules apply to them?

Variables and Data Types

- - What are primitive and reference data types in Java?
- - Declare variables of all primitive data types.
- - Write a program to input and display a student's name, age, and grade.

Type Casting

- - Differentiate between implicit and explicit casting.
- - Write a program to demonstrate both types of casting.
- - What happens when narrowing conversion fails?

Operators

- - Explain the difference between == and = operators.
- - Write a program using arithmetic and relational operators.

- - List and explain logical operators in Java.

Control Flow (Control Statements, Loops)

- - Write a program to find the largest of three numbers using if-else.
- - Demonstrate the use of switch-case.
- - Write a loop that prints numbers from 10 to 1.

Arrays

- - What are arrays in Java?
- - Write a program to find the sum of elements in an array.
- - Declare and initialize a 2D array and print its elements..

Problem Solving:

1. Find the Largest Element in an Array

Problem Statement:
Write a program to find the largest element in a given array of integers.

Input:
An array of integers, e.g., arr = [10, 5, 20, 8, 15]

Output:
The largest element in the array.
Example Output: 20

2. Find the Second Largest Element in an Array

Problem Statement:
Write a program to find the second largest element in a given array of integers.

Input:
An array of integers, e.g., `arr = [10, 5, 20, 8, 15]`

Output:
The second largest element in the array.
Example Output: `15`

3. Insert an Element at a Specific Position

Problem Statement:
Write a program to insert a new element into an array at a specific index.

Input:
Array: `[10, 20, 30, 40]`
Element to insert: `25`
Position: `2` (0-based indexing)

Output:
New array after insertion.
Example Output: `[10, 20, 25, 30, 40]`

4. Count Even and Odd Numbers in an Array

Problem Statement:
Write a program to count the number of even and odd numbers in an array.

Input:
Array: [1, 2, 3, 4, 5, 6]

Output:
Even count: 3, Odd count: 3

5. Find the Frequency of Each Element

Problem Statement:
Write a program to count the frequency of each element in an array.

Input:
Array: [10, 20, 20, 10, 30, 10]

Output:

10 occurs 3 times

20 occurs 2 times

30 occurs 1 time